

A Data-Driven Analysis of US Presidential Elections

Data Management & Predictive Modeling of US Elections (2000-2024)

Bourdouxhe Jessica Foidart Xavier

M1 Data Management - HEC Liège

January 2025

Final Project Presentation

Our Research Question:

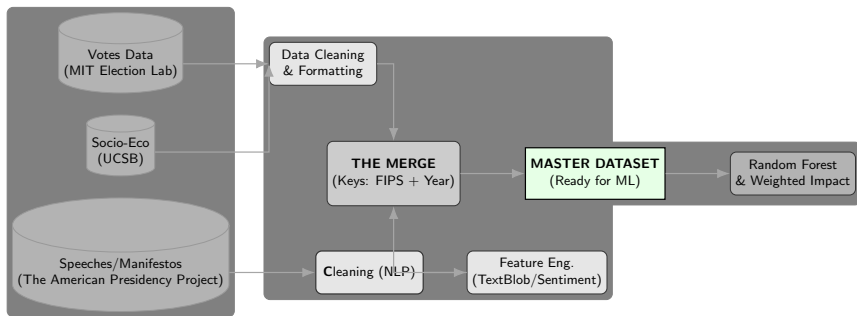
- Can macro-economic variables (education, age, ...) be used to predict election outcomes?
- Or has political rhetoric (NLP) and political identity taken over?

Objective: Build a robust Data Pipeline to integrate **Socio-Economic Data** with **Candidate Rhetoric Manifestos** to model the US County-level vote.

Data Engineering Pipeline

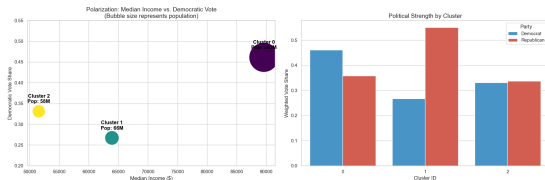
A modular ETL (Extract, Transform, Load) approach was implemented across 8 notebooks ('01_collection' to '08_report').

Our Data Pipeline is the architecture that allows us to start from heterogeneous raw files (both text and numbers) and channel them, via rigorous cleaning and merging steps, into a single final dataset ready for Machine Learning.



Descriptive Analysis: The Three Americas

Clustering (K-Means) revealed that the US is not binary (Red vs. Blue), but ternary.



- **Cluster 0 (Urban):** Safe Dem (+10 pts).
- **Cluster 1 (Rural):** Safe Rep (+28 pts).
- **Cluster 2 (The Gap):** High poverty, high diversity. **Margin j 1%.** This is the true battleground.

Modelling: The 2016 Structural Break

We trained Random Forest models to detect Feature Importance shifts.

Key Finding

The 2016 election amplified existing trends. Identity and Education solidified as the absolute dominant drivers, while economic variables lost significant predictive power.

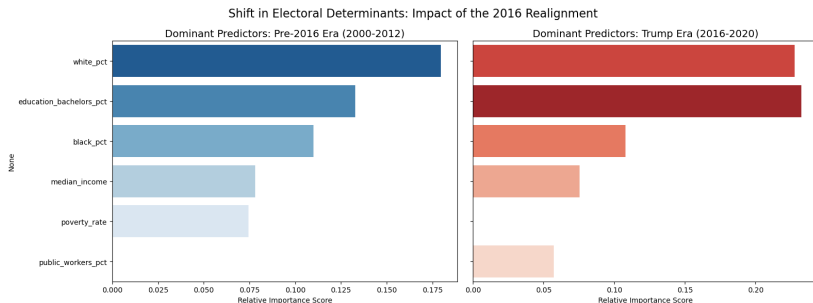
Pre-2016 Era:

- Driver: **Balanced Mix**
- Traditional economic variables (Median Income, Poverty Rate) were, alongside Identity variables, drivers of votes.

Post-2016 Era:

- Driver: **Identity & Education**
- **Education** and **% White** became even more dominant features in predictive power while economic variables lost predictive power.

Modelling: The 2016 Structural Break



NLP Analysis: The Limits of Rhetoric

Method: We analyzed the **Sentiment Polarity** of speeches and manifestos using TextBlob.

Year	Party	Speech	Platf.	Mean
2024	Democrat	0.184	0.107	0.146
	Republican	0.176	0.089	0.133
2020	Democrat	0.154	0.103	0.128
	Republican	0.157	-	0.157
2016	Democrat	0.171	0.130	0.151
	Republican	0.096	0.097	0.096
2012	Democrat	0.155	0.140	0.148
	Republican	0.192	0.107	0.149
2008	Democrat	0.152	0.122	0.137
	Republican	0.152	0.122	0.137

Table: Detailed Sentiment Scores

General Conclusion:

- **Stability:** Major parties maintain relatively stable sentiment scores over time.
- **No Correlation:** Unlike demographics, the "tone" of a campaign does not predict the vote share.
- **Identity > Mood:** The election is driven by structural identity, not by the speeches' sentiment.

**Scale: -1 (Neg) ↔ 0 (Neu) ↔ +1 (Pos)*

Conclusion & Critical Review

Main Takeaway

Our model indicates that Structural Identity (Education & Demographics) has solidified its dominance, while economic performance and campaign rhetoric have lost significant predictive power

Critical Review (Limitations):

- **Causality:** Our models show strong correlations, but we cannot claim causality without panel data tracking individual voters.
- **NLP Tool:** We used a lexical approach (TextBlob). Future work should implement **Transformers (BERT)** to capture sarcasm and context.
- **Socio-Economic Datas:** Feasibility Constraint for temporal socio economic datas.

Model Validation: Simulating Archetypes

We verified our machine learning model's (without NLP) logic by feeding it Synthetic Profiles (personas) to see how it reacts to specific socio-economic combinations.

1. The "Deep Red" Rural

Profile:

- **White:** 85%
- **Education:** Low (15%)
- **Income:** Moderate (\$45k)

Prediction:
REPUBLICAN
Conf: 96.1%

2. The "Blue" Metropolis

Profile:

- **White:** 45% (Diverse)
- **Education:** High (55%)
- **Income:** High (\$85k)

Prediction:
DEMOCRAT
Conf: 81.7%

3. The Suburban Stress-Test

Conflict:

- **Income:** \$95k (High)
- **White:** 80% (High)
- **Education:** 40% (High)

Outcome:
REPUBLICAN
Conf: 72.8%

Thank You.

Q&A